

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1102

COURSE OUTCOME COVERED

CO4: Implement object-oriented system layers using design principles and patterns, and integrate access and view layers with OODBMS (BL3)
Assessment Tool Weightage: 30%

QUESTIONS: 15

Total Marks:30

Annexure A

CLASS TEST

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

20/08/2026

CO4-AT-1

Delisha A K

192511307

slot D

CSA1110

Q6. Three Tier - Architecture

→ Object Oriented application can be divided into three separate layers. They are;

- (i) Presentation layer
- (ii) Business layer / Business logic layer.
- (iii) Data layer

(i) Presentation layer: It is the layer responsible for giving input and getting and presenting output to the user.

(ii) Business layer: It has two works + validation + processing

→ It checks if the input given by the user is valid and processes the data according to the business rules to be followed.

(iii) Data layer: It is responsible for storing, retrieving, and all works related to data.

workflow:

presentation layer → Business layer → Data layer →

eg: E-commerce site: In this the Presentation layer is responsible for taking the input from the user after that Business layer checks the valid input and process it and after finishing it integrates with the data layer which stores the order given by the user in the presentation layer and all the information related to it.

Q7. Business Layer Logic

- It is one of the three layers in three-layer architecture in OOP (object-oriented programming).
- It is responsible for checking the business rules of that particular platform and act according to it in the processing unit.
- It has two main parts: * validation
* processing.

* Validation:

- It is responsible for checking if the input given the presentation layer is valid.
- It also check the quantity, availability of the request.

* Processing:

- It is responsible for performing operations according to the business logic and rules.

∴ It keeps business rules separate from the user interface, making the system more organised, maintainable, scalable and reusable.

eg: In an online shopping system the business layer is responsible for getting logics and requests from the presentation layer and work with data layer to organise and manage the data's received.

Cohesion and Coupling:

classes: classes are the blue print of the objects

objects: These are real-world entities.

analysis: It is the process in which we identify the objects, classes and attributes and all the required information of the designing part.

designing: It is the part in which we identify how objects are related, how the design will integrate for the implementation part and also find how message passes between objects.

Cohesion:

→ It is the representation of how closely the objects are related in a class.

Coupling:

→ It refers to the dependency level between different modules or classes.

In a good object-oriented design we prefer to is
high cohesion ↑ and low coupling ↓

eg: A payment scribe should mainly handle only payment related thing not use interface and processing.

Importance in OOP

- high cohesion makes classes easier to understand and reuse
- low coupling makes them easier to test and do changes and modification.

- Q4. Encapsulation and data hiding
- classes: classes are the blueprint, which stores data and behavior of the particular module.
 - Object: objects are the real-world entities which is implemented.
 - OOP: object oriented programming uses objects to the programming for better scalability and maintainability.

The four pillars in OOP are;

- (i) Encapsulation
- (ii) Abstraction
- (iii) Inheritance
- (iv) Polymorphism

(i) Encapsulation :

- It is one of the main pillar in OOP
- It is responsible for the binding data and behavior, implementation into single unit
- It is responsible for modular activity.
- It restricts the direct access of internal data access using methods().

Data hiding:

- It is used to hide the secure and sensitive data from the users using access specifiers.

```
eg: public static void main(String[] args)
    {
        private class Demo( )
    }
```

layered system for online banking system.

→ an online banking system can be designed using different layers:

1) Presentation layer

→ provides the interface through which clients interact to see their information

2) Application layer

→ used for coordinating banking operations such as money transfer, deposit.

3) Business logic layer:

→ contains rules such as checking amount balance and validating transactions.

4) data layer

→ stores, client, account and transaction details in database

5) Infrastructure layer:

→ Provides technical information and organising connectivity to external layer and API.

Importance of it in OOP

→ using different layers we get;

- * security
- * scalability
- * maintainability

→ design patterns used are *

* MVC - User interface

* DAO - data access layer